

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Lab Task # 07

COURSE:

Artificial Intelligence Lab_CS-202L

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Task # 01

```
from collections import deque

graph = {
    0: [1, 3, 4],
    1: [2],
    2: [],
    3: [5],
    4: [5],
    5: []
}

def bfs(start):
    visited = []
    queue = deque([start])

    while queue:
        node = queue.popleft()
        if node not in visited:
            visited.append(node)
            queue.extend(graph[node])
    return visited

print("BFS:", bfs(0))
```

Output

```
BFS: [0, 1, 3, 4, 2, 5]
```

Task # 02

```
graph = {
    1: [2, 7],
    2: [3, 6],
    3: [4, 5],
    4: [],
    5: [],
    6: [],
    7: [8, 9],
    8: [],
    9: [10, 11],
    10: [],
    11: []
}

def dfs(start, visited=None):
    if visited is None:
        visited = []
    visited.append(start)

    for neighbor in graph[start]:
        if neighbor not in visited:
            dfs(neighbor, visited)
    return visited

print("DFS:", dfs(1))
```

Output

```
DFS: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
```